

PROVISIONAL PATENT APPLICATION

Title: SPECTRE N0L: Asynchronous Frequency-Activated Self-Timed Inter-Module Communication Engine Where Frequency-Encoded Pulses Serve Simultaneously as Data Carriers and Handshake Signals, Eliminating the Global Clock Tree

COVER SHEET

Application Type: Provisional Patent Application (35 U.S.C. § 111(b))

Title of Invention: Frequency-Activated Asynchronous Self-Timed Processing Architecture in Which a Shared Frequency Vocabulary Serves as Both Data Encoding Alphabet and Inter-Module Handshake Protocol, With Delay-Insensitive Cross-Platform Synchronization and Zero-Idle-Power Operation

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Parent Application Reference: This application is a provisional extension of SES-003-SPECTRE, filed May 4, 2026. It extends the frequency-domain encoding paradigm of SPECTRE to the inter-module handshake and system timing layer, and claims priority to the parent project SPECTRE priority date of April 25, 2026.

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FIELD OF THE INVENTION

This invention relates to asynchronous digital logic, self-timed inter-module communication, and frequency-domain data processing systems implemented in reconfigurable hardware (FPGA) and associated digital logic. More particularly, this invention relates to an architecture in which frequency-encoded pulses — derived from the MIDI-mapped frequency vocabulary established in the parent SPECTRE system — serve simultaneously as data carriers and as inter-module handshake signals, collapsing the data path and the control path into a single physical signal and eliminating the global clock tree from the inter-module communication layer.

BACKGROUND

1.1 The Global Clock Constraint

Every conventional synchronous digital system — including all current commercial FPGA designs — operates under a global clock tree. The clock tree distributes a periodic timing signal to every flip-flop in the design. All state transitions occur on clock edges. All combinational logic must settle within one clock period (the critical path). The clock period is set by the slowest path in the design.

This architecture imposes three fundamental costs:

Dead time: Logic that completes its computation in 0.3 ns of a 1.67 ns clock period (600 MHz) sits idle for the remaining 1.37 ns, waiting for the next clock edge. This dead time can represent 30–80% of total potential throughput depending on the distribution of path delays across the design.

Clock power: The clock tree toggles continuously — every register, every clock buffer, every clock spine — at full frequency, even when no data is moving. Dynamic power consumption is dominated by clock tree switching in large synchronous designs, often representing 30–50% of total chip power.

Clock skew as scaling limit: As the number of connected processing nodes grows, maintaining sub-nanosecond clock skew across all nodes becomes exponentially more difficult. At board-to-board and system-to-system scales, a shared global clock is physically impossible without specialized clock distribution hardware (e.g., White Rabbit PTP). Even with such hardware, the shared clock remains a single point of failure and a fundamental scaling ceiling.

1.2 Asynchronous and Self-Timed Logic

Asynchronous digital logic — logic that operates without a global clock — has been studied since the 1950s. Key paradigms include:

- **Muller handshake circuits:** REQ/ACK two-phase or four-phase handshake protocols between adjacent logic stages.
- **Null Convention Logic (NCL):** Data is encoded in three states — NULL (no data), DATA0 (logic zero), DATA1 (logic one). Threshold gates (TH_{mn}) fire only when a complete set of input states has arrived. The NULL state between data words separates successive computations and prevents wave collisions.
- **Wave pipelining:** Multiple data waves traverse the same combinational logic simultaneously. Registers between pipeline stages are eliminated. Throughput improvement of up to 5.7× over synchronous pipelines has been demonstrated in equalized-path implementations.
- **Delay-insensitive (DI) circuits:** Circuits whose correct operation is guaranteed regardless of the propagation delays of individual gates and wires, as long as delays are finite and positive. DI circuits automatically adapt to silicon variation, temperature, and supply voltage.

Despite decades of research, asynchronous logic has seen limited commercial FPGA adoption because: (a) standard EDA tools (Vivado, Quartus) are designed for synchronous analysis; (b) the REQ/ACK handshake signals require additional wires separate from the data path; and (c) asynchronous designs are difficult to verify with standard simulation flows.

1.3 The SPECTRE Frequency Vocabulary

The parent invention SPECTRE (SES-003) established a 128-entry bijective frequency vocabulary: the MIDI equal-temperament pitch standard maps each 7-bit data symbol to a unique acoustic frequency between 8.18 Hz and 12,543.85 Hz. This vocabulary has been validated as a lossless data encoding alphabet across nine FPGA device families.

A critical empirical observation from SPECTRE validation: **cross-board timing variance is less than 3 nanoseconds** across nine device families spanning 28nm through 7nm-equivalent process nodes. This is a remarkable result for a system in which the transmitting and receiving boards run entirely independent clocks with no shared synchronization signal. The mechanism is that SPECTRE's Goertzel-based decoder is inherently delay-insensitive: it detects which frequency is present in a 480-sample window, not when (relative to any absolute time reference) that frequency arrives. The frequency measurement is invariant to clock drift, temperature variation, and silicon speed grade.

This observation establishes that SPECTRE's frequency vocabulary already implements delay-insensitive inter-board communication as an emergent property of its encoding paradigm

— without having been designed to do so explicitly. SPECTRE N0L makes this emergent property explicit, intentional, and architecturally foundational.

1.4 The Uncaptured Opportunity

No prior art is known that:

1. Collapses the data encoding signal and the inter-module handshake signal into a single physical frequency-encoded pulse.
 2. Applies the MIDI-mapped acoustic frequency vocabulary as a system-wide inter-module communication protocol (not merely a data encoding alphabet).
 3. Achieves delay-insensitive inter-board synchronization as a natural consequence of frequency-domain encoding rather than as an engineered clock distribution solution.
 4. Enables unlimited board-count scaling by replacing clock-based synchronization with frequency-arrival-based wake-up.
 5. Achieves zero-idle-power operation in reconfigurable hardware by making module switching activity exactly proportional to incoming frequency wave rate.
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SUMMARY OF THE INVENTION

SPECTRE N0L (No-clock / No Limits) is an asynchronous, self-timed, frequency-activated inter-module communication architecture that extends the SPECTRE frequency vocabulary from a data encoding layer to a universal system protocol layer. In SPECTRE N0L, every module in the processing pipeline — and every board in a multi-board system — communicates exclusively through SPECTRE frequency pulses. There is no global clock. There is no shared synchronization signal. There is only: **frequency arrives** → **module activates** → **processes** → **emits next frequency** → **returns to dark state**.

The central innovation is the collapse of data and control into a single physical signal. In conventional asynchronous logic (e.g., NCL, Muller handshake circuits), data wires and REQ/ACK handshake wires are separate. A data word is accompanied by a separate REQ signal asserting "data is present," and a separate ACK signal returning "data has been consumed." In SPECTRE N0L, the REQ signal is the data: the arrival of a frequency pulse at a module's input simultaneously carries the 7-bit data payload AND asserts the request for the receiving module to begin processing. The receiving module's completion of Goertzel frequency detection simultaneously recovers the data payload AND generates the acknowledgment that the transmission is complete.

The NULL state in SPECTRE N0L corresponds directly to the NULL state in Null Convention Logic: the absence of any SPECTRE frequency pulse. A module in the NULL state is consuming only static leakage power — no dynamic switching, no clock toggling, no idle

current. A module transitions from NULL to active in the same instant the first sample of an incoming frequency pulse enters its Goertzel accumulator. This is the physical implementation of zero-idle-power operation.

The wave pipeline emerges naturally: because each module activates independently on frequency arrival and completes independently on Goertzel detection, multiple frequency waves can be in transit through the pipeline simultaneously — one being detected by Module N while Module N-1 is already emitting the next frequency and Module N+1 is already staging its Goertzel accumulator. The pipeline stages are separated not by registers clocked on a shared edge, but by the physical propagation delay of the frequency pulse through the transmission medium.

Infinite board scaling follows directly: adding a board to the system requires only connecting it to the frequency transmission medium. No clock distribution infrastructure is required. No PLL lock sequence. No clock skew budget reallocation. The new board wakes up when a frequency pulse arrives, processes it, and emits the next frequency. The system scales by the laws of physics, not by the constraints of clock tree engineering.

BRIEF DESCRIPTION OF DRAWINGS

(Informal drawings — formal drawings to accompany non-provisional application)

Figure 1 — SPECTRE vs. SPECTRE N0L Protocol Layer Comparison Side-by-side: Left shows SPECTRE (SES-003) with synchronous RTL inside each board and frequency encoding at the board-to-board interface only. Right shows SPECTRE N0L with frequency-activated handshake at every inter-module interface, including intra-board. Clock tree is shown present in SPECTRE, absent in N0L. Frequency wave arrows replace clock distribution spines.

Figure 2 — Data+Control Collapse Diagram Conventional asynchronous: three wires per interface (DATA, REQ, ACK). NCL: DATA bus + separate REQ/ACK handshake rails. SPECTRE N0L: single differential pair carries frequency-encoded pulse that simultaneously IS the data, IS the REQ, and whose Goertzel detection completion IS the ACK. Wire count reduction: 3-wire interface to 1-wire (differential pair) interface.

Figure 3 — NULL State Power Consumption Model Two curves: (a) synchronous clocked module — constant dynamic power $P = \alpha \cdot C \cdot V^2 \cdot f$ regardless of data rate; (b) SPECTRE N0L module — dynamic power $P = 0$ when no frequency present, $P \propto f_{\text{wave}}$ when frequency wave active. Crossover point shown: N0L is more power-efficient than synchronous at any data rate below clock frequency.

Figure 4 — Wave Pipeline in SPECTRE N0L Four modules (M1–M4) in pipeline. Three frequency waves (W1, W2, W3) shown simultaneously in transit: W1 being detected by M2 Goertzel while W2 is in the transmission medium between M1 and M2 while W3 is being emitted by M1. No registers between stages. Wave separation maintained by physical propagation delay.

Figure 5 — Infinite Board Scaling Architecture Linear chain: Board 1 → Board 2 → ... → Board N. Each board connected by LVDS differential pair carrying SPECTRE frequency pulses. No clock distribution. Board N receives frequency, Board N-1 receives frequency simultaneously. Each board independently Goertzel-detects its incoming frequency and emits the next. Adding Board N+1 requires only connecting one LVDS pair.

Figure 6 — Goertzel Completion as ACK Timing Diagram Waveform: frequency_pulse_in (differential signal), goertzel_accumulator (ramps over 480 samples), max_power_bin (asserts at end of window = data recovered), ack_implicit (conceptual signal: max_power_bin assertion IS the ACK, no separate wire). Compare to NCL: explicit ACK wire shown for contrast.

Figure 7 — SPECTRE N0L Module FSM States: ST_NULL (dark, zero dynamic power) → ST_ACCUMULATE (Goertzel running, 480 samples) → ST_DETECT (max bin selected, data recovered) → ST_EMIT (next frequency pulse transmitted) → ST_NULL. No clock edge triggers any transition — transitions are triggered by: frequency_present (NULL→ACCUMULATE), accumulator_full (ACCUMULATE→DETECT), detection_complete (DETECT→EMIT), emission_complete (EMIT→NULL).

Figure 8 — SES System Fully Realized in N0L Complete pipeline: Shimmers oscillator (frequency source) → Eidolon Dragoncache (all inter-tier interfaces replaced with SPECTRE N0L frequency handshakes) → SPECTRE N0L multi-board pipeline → Storage / Analysis. Every arrow in the diagram is a SPECTRE frequency pulse. Every module wakes on frequency arrival. One language. No clocks.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

4.1 The Foundational Principle: One Signal, Three Roles

In SPECTRE N0L, a single frequency-encoded pulse simultaneously performs three roles that are implemented as three separate signals in all prior art:

Role 1 — Data carrier: The frequency of the pulse identifies the 7-bit data payload via the MIDI-mapped bijective LUT established in SPECTRE (SES-003). A pulse at 440 Hz (MIDI note

69) carries the 7-bit symbol 0x45. A pulse at 261.63 Hz (MIDI note 60, middle C) carries 0x3C. The complete 128-entry mapping is inherited from SPECTRE with no modification.

Role 2 — Request (REQ) signal: The presence of any SPECTRE frequency pulse at a module's input — detectable within the first few samples of the Goertzel accumulator as rising spectral power — implicitly asserts REQ to the receiving module. The module transitions from ST_NULL to ST_ACCUMULATE without any separate wire being asserted and without waiting for a clock edge.

Role 3 — Implicit acknowledgment generation: The receiving module's completion of Goertzel detection (ST_DETECT) is the implicit ACK. The transmitting module considers its transmission acknowledged when the next frequency pulse arrives from downstream — either directly as a separate feedback frequency or inferred from the pipeline's equalized-path timing. In a unidirectional pipeline (the preferred embodiment), the ACK is implicit in the pipeline flow and no explicit acknowledgment wire is required.

This collapse from three signals to one is the primary engineering innovation of SPECTRE N0L. It reduces inter-module interface complexity to a single differential pair (LVDS or equivalent) per directed connection.

4.2 Novel Contribution 1 — Frequency Pulse as Simultaneous Data and Handshake

Prior art interface (NCL/Muller):

Module A → Module B:

DATA[6:0] : 7 wires (data payload)

REQ : 1 wire (data valid assertion)

ACK (B→A) : 1 wire (consumption acknowledgment)

Total: 9 wires minimum per interface

SPECTRE N0L interface:

Module A → Module B:

FREQ_P : 1 wire (differential positive — frequency-encoded pulse)

FREQ_N : 1 wire (differential negative)

Total: 2 wires per interface

The 7-bit data payload is recovered by Goertzel detection of the `FREQ_P/FREQ_N` signal. The REQ is implicit in the pulse's presence. The ACK is implicit in the pipeline's flow. Interface complexity drops from 9 wires to 2 wires per directed connection — an 77.8% reduction in inter-module wiring.

At system scale — N modules with M inter-module connections — the wiring reduction is $M \times 7$ wires. For the 13-module Dragoncache implementation (Eidolon, SES-002), this represents a reduction from approximately 117 inter-module data+handshake wires to 26 differential pair wires.

4.3 Novel Contribution 2 — Delay-Insensitive Cross-Platform Synchronization as Designed Property

The SPECTRE parent system (SES-003) demonstrated empirically that cross-board timing variance is less than 3 nanoseconds across nine FPGA device families with no shared clock. SPECTRE N0L makes this property architectural and intentional rather than emergent.

The mechanism: Goertzel frequency detection is mathematically a measurement of spectral power at a target frequency over N samples. The measurement result — the detected MIDI note — is independent of: the absolute time at which the frequency pulse arrived, the phase offset of the receiving module's sampling clock relative to the transmitting module's clock, the temperature and supply voltage of the receiving silicon (which affect gate delays and therefore sampling clock accuracy), and the speed grade of the device family.

The only requirement for correct detection is that $N = 480$ samples are accumulated at 96,000 samples per second. The sample clock frequency must be stable over the 5 ms window duration — a requirement met by any FPGA PLL with standard jitter specifications.

This property is delay-insensitive by the formal DI definition: the system's logical correctness is guaranteed for any finite positive propagation delay. Adding a board to the pipeline introduces additional propagation delay but does not change the correctness of any module's Goertzel detection. The system adapts automatically.

4.4 Novel Contribution 3 — Zero-Idle-Power Operation

In a synchronous clocked system, dynamic power consumption is:

$$P_{\text{dynamic}} = \alpha \times C \times V^2 \times f_{\text{clock}}$$

where α is the switching activity factor (fraction of flip-flops and logic gates that switch per clock cycle). Even when no data is being processed, $\alpha > 0$ because the clock tree itself toggles every

flip-flop's clock input at every clock edge. In large designs, the clock tree alone can represent $\alpha \approx 0.3\text{--}0.5$ of total switching activity.

In a SPECTRE N0L module, there is no clock tree. When no frequency pulse is present (ST_NULL state), the only power consumed is static leakage through transistor gate oxides. Dynamic switching activity is zero. Dynamic power is zero. The module is "dark."

When a frequency pulse arrives, the Goertzel accumulator begins switching. Dynamic power scales linearly with the incoming pulse frequency:

$$P_{\text{dynamic_N0L}} = \alpha_{\text{goertzel}} \times C \times V^2 \times f_{\text{pulse_rate}}$$

where $f_{\text{pulse_rate}}$ is the rate of incoming frequency pulses (at most 96,000 per second for a 480-sample window at 96 kHz) and α_{goertzel} is the switching activity of the Goertzel accumulator only.

The zero-idle-power property enables a new class of system architecture: a pipeline of N boards can remain fully dark until a single triggering frequency pulse is injected at Board 1. The pulse propagates through the pipeline, activating each board only during its active processing window, then returning each board to the dark state. Total energy consumed per data unit is bounded and proportional to data volume — not to wall-clock time.

4.5 Novel Contribution 4 — Wave Pipeline Without Registers

In a standard synchronous pipeline, N pipeline stages require N-1 register stages (flip-flop banks) to hold data at stage boundaries while downstream stages complete their work. Each register stage consumes FPGA flip-flop resources and introduces one clock cycle of latency.

In SPECTRE N0L, register stages between pipeline modules are eliminated. Multiple frequency waves can be in transit through the pipeline simultaneously, separated by the physical propagation delay of the transmission medium (approximately 5 ns per meter for electrical signals in PCB trace or coaxial cable; approximately 3.3 ns per meter in fiber optic).

The equalized-path condition: For wave pipelining to be correct, the minimum and maximum propagation delays through each pipeline stage must be equalized — that is, the difference between the fastest and slowest path through any stage must be less than one wave spacing. In SPECTRE N0L, each module's processing time is deterministic and equal: 480 Goertzel accumulation cycles + detection + emission = a fixed number of clock cycles at the local sampling clock. This determinism is an inherent property of the Goertzel algorithm applied to a fixed-length window. The equalized-path condition is satisfied by design.

Research in conventional wave pipelining demonstrates throughput improvements of up to $5.7\times$ over synchronous pipelines with equivalent logic depth. SPECTRE N0L achieves this

improvement without any of the specialized EDA tooling required for conventional wave pipeline design, because the pipeline stages are separated by physical transmission medium (wire/fiber) rather than by on-chip combinational logic paths.

4.6 Novel Contribution 5 — Universal Frequency Protocol: Every Module Speaks SPECTRE

In the parent SPECTRE system (SES-003), the frequency vocabulary is used at one layer: the data encoding interface between the encoder and the WAV storage / Goertzel decoder. Modules internal to the FPGA communicate via standard synchronous RTL signals (`std_logic_vector`, valid/ready handshake, AXI4-Lite, etc.).

In SPECTRE N0L, the MIDI-mapped 128-frequency vocabulary is the universal protocol for ALL inter-module communication. Every module in the system — regardless of its function (oscillator, hub, `extent_tree`, `stack_reader`, CRC engine, PCIe interface) — has a SPECTRE N0L interface: it receives frequency pulses as input and emits frequency pulses as output. The module's function determines which frequency it emits in response to which incoming frequency, but the physical interface is identical for every module.

The architectural consequence: Every module and every board in the system is natively interoperable. A board does not need to know whether it is communicating with an oscillator module or a CRC engine module. It receives a frequency, processes it, emits the next frequency. The system can be reconfigured by changing the mapping between incoming and outgoing frequencies at each module — a software operation on the frequency-routing table — without any hardware change.

The "fleet" scaling property: A fleet of N boards connected by LVDS differential pairs all running SPECTRE N0L are natively interoperable from the moment they are powered. Board N does not need to be configured to communicate with Board M — both already speak SPECTRE. Adding a board is equivalent to adding a native speaker to a conversation: no translation layer required.

4.7 Novel Contribution 6 — The NULL State as Architecture

In Null Convention Logic (NCL), the NULL state is a data encoding level that separates successive data words in a bundled data channel. It is a protocol artifact — a separator that must be explicitly generated and detected.

In SPECTRE N0L, the NULL state is the physical absence of any frequency pulse on the transmission medium. It requires no explicit generation (a module simply stops transmitting), no explicit detection (a module simply observes that no Goertzel bin has sufficient power), and no additional protocol overhead. The NULL state is free: it costs nothing to generate, nothing to detect, and nothing to transmit.

This property — NULL as the absence of signal rather than as an explicit signal level — is unique to frequency-domain encoding. In binary encoding, silence (no signal) is ambiguous: it could mean NULL, or it could mean a transmission error, or it could mean the line is idle. In SPECTRE N0L, silence has unambiguous meaning: no data is present. A module in a silent-input state remains in ST_NULL. There is no ambiguity.

The NULL state in SPECTRE N0L therefore simultaneously implements: wave separation (prevents wave collisions in the wave pipeline), idle detection (distinguishes no-data from data), and power gating (the module's switching activity drops to zero in the NULL state automatically, without any power management logic).

4.8 Novel Contribution 7 — Raw LVDS Pin Interface: Bypassing Transceiver Clock Recovery

Standard FPGA high-speed transceivers (e.g., AMD GTY, GTM) incorporate Phase-Locked Loops (PLLs) and Clock Data Recovery (CDR) circuits that re-synchronize received serial data into the local clock domain. This is the correct behavior for protocols that require synchronous processing (PCIe, Ethernet, JESD204B). For SPECTRE N0L, it is exactly the wrong behavior: CDR would force the received frequency pulse back into a clocked domain, destroying the asynchronous self-timed property.

SPECTRE N0L bypasses high-speed transceivers and uses raw LVDS differential I/O pins with IBUFDS (input differential buffer) at the receiver and OBUFDS (output differential buffer) at the transmitter. The received differential signal feeds directly into the Goertzel accumulator's sample input. No CDR, no PLL lock sequence, no 8b/10b framing overhead.

LVDS on UltraScale+ HR I/O banks supports signaling rates up to approximately 1.6 Gbps in raw mode. For SPECTRE N0L's 96 kHz sample rate (well below this limit), LVDS provides ample bandwidth with sub-nanosecond signal integrity on PCB traces and with standard SFP/optical transceivers for inter-board connections.

The Xilinx IDELAY/ODELAY primitives (IDELAYE3/ODELAYE3 on UltraScale+) provide per-pin delay adjustment with 2 ps resolution over a 1.25 ns range. These primitives enable per-board wave pipeline deskewing: if Board M's Goertzel window completes slightly early or late relative to Board M+1's accumulation start, ODELAY at Board M's output and IDELAY at Board M+1's input can be tuned to compensate. This is the hardware equivalent of equalizing the wave pipeline paths across a multi-board system.

4.9 Integration as the Culminating Architecture of the SES System

SPECTRE N0L is the culminating expression of the Shimmers · Eidolon · SPECTRE system. When all three components are realized in N0L:

Shimmers (SES-001) in N0L context: The oscillator engine generates the first frequency in the pipeline — the source wave. The DDS synthesizer emits a MIDI-frequency pulse whose frequency encodes the first data symbol. This pulse is the genesis of the entire pipeline. No clock starts the system — the first frequency pulse, self-generated by the oscillator, starts everything.

Eidolon (SES-002) in N0L context: Every inter-tier interface in the Dragoncache — Inception→Hub, Hub→stack_reader, stack_reader→extent_tree, Conductor FSM state transitions — is replaced with SPECTRE frequency handshakes. The Conductor FSM's state transitions are triggered by frequency arrival, not by clock edges. Chamber A/B alternation is triggered by a specific designated "chamber switch" MIDI note. CRC64 seal assertion is carried as a specific "seal valid" frequency. The entire IPC backbone speaks SPECTRE.

SPECTRE (SES-003) in N0L context: The wav_encoder and Goertzel decoder are the output and recovery interfaces — unchanged from SES-003. SPECTRE N0L feeds the wav_encoder a pre-processed, pipeline-validated frequency stream that has already passed through the Eidolon integrity gate.

The emergent system property of full N0L integration: In the fully realized SES N0L system, there is no global clock, no shared synchronization signal, and no inter-module wiring beyond LVDS differential pairs. The system boots when the first frequency pulse is emitted. It processes data at the rate frequency pulses arrive. It consumes zero power when no frequency pulses are present. It scales to any number of boards by connecting additional LVDS pairs. The language of every signal in the system, at every layer, is SPECTRE.

This is not the end of the beginning. It is the beginning of what comes next.

CLAIMS

(Informal claims for provisional filing — formal claims to be developed with patent attorney for non-provisional application)

Claim 1. An asynchronous self-timed inter-module communication system comprising: a transmitting module that encodes a data symbol as a frequency-encoded pulse by selecting a pulse frequency from a predefined bijective frequency vocabulary based on the data symbol value; a transmission medium that carries the frequency-encoded pulse from the transmitting module to a receiving module; and a receiving module that recovers the data symbol from the received pulse by spectral frequency detection, wherein the presence of the frequency-encoded pulse simultaneously constitutes the data payload, the request signal to the receiving module to

begin processing, and the trigger event that transitions the receiving module from an idle state to an active processing state, without requiring a separate request wire or a shared global clock.

Claim 2. The system of Claim 1, wherein the receiving module's completion of spectral frequency detection constitutes an implicit acknowledgment that the data symbol has been received, without requiring a separate acknowledgment wire.

Claim 3. The system of Claim 1, wherein the bijective frequency vocabulary consists of the 128 pitches of the MIDI equal-temperament standard, mapping each 7-bit data symbol to a unique frequency between 8.176 Hz and 12,543.85 Hz.

Claim 4. The system of Claim 1, wherein spectral frequency detection is performed by the Goertzel algorithm operating on a fixed-length sample window, and wherein the detection result is delay-insensitive: correct for any finite positive propagation delay between the transmitting module and the receiving module, without requiring clock synchronization between the two modules.

Claim 5. The system of Claim 1, wherein the receiving module consumes zero dynamic power when no frequency-encoded pulse is present on the transmission medium, and wherein dynamic power consumption is proportional to the rate of incoming frequency-encoded pulses.

Claim 6. The system of Claim 1, wherein a plurality of transmitting and receiving modules are interconnected in a pipeline, wherein multiple frequency-encoded pulses from successive data symbols are simultaneously in transit through the pipeline, separated by the physical propagation delay of the transmission medium, constituting a wave pipeline without register stages between pipeline modules.

Claim 7. A multi-board processing system comprising a plurality of processing boards, each board implementing the receiving module of Claim 1, wherein the boards are interconnected by differential pair transmission media carrying frequency-encoded pulses, and wherein no shared global clock signal is distributed between boards, and wherein additional boards are added to the system without clock distribution reconfiguration.

Claim 8. The system of Claim 7, wherein cross-board timing variance is less than 3 nanoseconds across boards of differing semiconductor process nodes, device families, operating temperatures, and supply voltages, as a consequence of the delay-insensitive property of frequency-domain detection.

Claim 9. A method of eliminating the global clock tree from an inter-module communication system comprising: defining a shared frequency vocabulary as the universal protocol for all inter-module communication; encoding every inter-module data transfer and every inter-module control signal as a frequency-encoded pulse selected from the shared frequency vocabulary; detecting inter-module communication events by spectral frequency analysis of incoming

pulses; and triggering module state transitions on frequency detection completion rather than on clock edges.

Claim 10. The method of Claim 9, wherein a designated frequency in the shared vocabulary encodes each distinct control event — including pipeline stage advance, chamber alternation, integrity seal assertion, and lane switch request — such that data and control signals share a common physical interface and a common detection mechanism.

Claim 11. An integrated self-timed processing system comprising: an oscillator module that generates the first frequency-encoded pulse in a pipeline from an internally derived frequency, without requiring an external trigger; one or more intermediate processing modules that each receive a frequency-encoded pulse, perform a deterministic processing operation, and emit the next frequency-encoded pulse encoding the result; and an output module that receives the final frequency-encoded pulse and writes the recovered data symbol to a storage medium; wherein no global clock is shared among any of the modules, and wherein the pipeline is initiated solely by the first frequency pulse emitted by the oscillator module.

Claim 12. The system of Claim 11, wherein the system is the terminal realization of an integrated architecture comprising a frequency oscillator engine, a three-tiered inter-process communication engine, and a frequency-domain data transceiver, and wherein all inter-module interfaces at all layers of all three components operate exclusively through frequency-encoded pulses from a shared MIDI-mapped frequency vocabulary, constituting a unified frequency-domain processing system in which every module and every board is natively interoperable without translation layers or shared clock distribution.

ABSTRACT

SPECTRE N0L (No-clock / No Limits) is an asynchronous self-timed inter-module communication architecture extending the SPECTRE frequency-domain encoding paradigm (SES-003) from the data layer to the system timing and control layer. In SPECTRE N0L, frequency-encoded pulses from the MIDI-mapped 128-frequency vocabulary serve simultaneously as data carriers, inter-module request signals, and activation triggers, eliminating the need for separate handshake wires and eliminating the global clock tree from the inter-module communication layer. The receiving module's spectral frequency detection completion constitutes an implicit acknowledgment. Modules consume zero dynamic power when no frequency pulse is present. Multiple data waves propagate through the pipeline simultaneously without register stages. Cross-board timing variance below 3 nanoseconds is achieved across all device families as an inherent consequence of Goertzel detection's delay-insensitive property, without shared clock distribution. Unlimited board scaling is enabled: each board wakes on frequency arrival, processes, emits the next frequency, and returns to dark state. When applied to the complete Shimmers · Eidolon · SPECTRE integrated system,

every inter-module interface at every layer operates exclusively through SPECTRE frequency pulses — one language, no clocks, no limits.

INVENTOR DECLARATION

I hereby declare that I am the sole inventor of the subject matter claimed in this provisional patent application; that I believe I am the original and first inventor of the invention; that I have reviewed and understand the contents of this application; and that I acknowledge my duty to disclose information material to patentability.

The concept of using frequency-encoded pulses simultaneously as data carriers and inter-module handshake signals, and the resulting architectural properties (zero-idle power, unlimited board scaling, delay-insensitive synchronization, wave pipeline without registers), were conceived by me. All artificial intelligence tools used in the development and documentation of this invention were used as instruments under my direction. They made no independent inventive contribution.

Inventor: Scott Slater **Date:** May 4, 2026

This provisional patent application establishes a priority date for the invention described herein. It extends the parent provisional SES-003-SPECTRE filed May 4, 2026, and claims priority to the parent project SPECTRE priority date of April 25, 2026. A non-provisional patent application must be filed within twelve (12) months of the filing date of this provisional pursuant to 35 U.S.C. § 119(e).

Filed under: SES-004-SPECTRE-N0L | Docket: Shimmers · Eidolon · SPECTRE · N0L | The end of the beginning.